

# Homework 4 — Function families

Session 4 · Polynomials, exponentials, logarithms, and trigonometric models

**HOW TO ANSWER** Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

**BEFORE YOU START** Conditions are part of an answer. Before using a logarithm, state which inputs are allowed. When you compare growth, say *for sufficiently large  $n$* ; one numerical example does not prove a growth claim.

## Part A · Mechanics

### 1 POLYNOMIAL SHAPE

Let  $p(x) = -3x^4 + 2x - 7$ .

- State the degree and leading coefficient of  $p$ .
- Describe what the graph does at its left and right ends.

Give one sentence explaining why the lower-power terms do not change your answer to part (b).

**SOLUTION** The degree is 4 and the leading coefficient is  $-3$ . The degree is even and the leading coefficient is negative, so the graph goes down at both ends:  $p(x) \rightarrow -\infty$  as  $x \rightarrow -\infty$  and as  $x \rightarrow \infty$ . When  $|x|$  is large, the leading term  $-3x^4$  is much larger in size than  $2x - 7$ , so it determines the end behaviour.

### 2 QUADRATIC TRANSFORMATION

Start with  $f(x) = x^2$ . The function

$$g(x) = -(x - 2)^2 + 4$$

is made from  $f$ .

- Describe the shift and reflection.
- Give the vertex and find the real roots of  $g$ .

**SOLUTION** Move the graph of  $y = x^2$  right by 2, reflect it in the  $x$ -axis, then move it up by 4. Its vertex is  $(2, 4)$ . For roots,  $-(x - 2)^2 + 4 = 0 \iff (x - 2)^2 = 4 \iff x - 2 = \pm 2$ , so  $x = 0$  or  $x = 4$ .

### 3 EXPONENTIAL EQUATION

Solve  $5^{2x-1} = 125$ . Rewrite **125** with base **5**, then state why you may set the exponents equal.

**SOLUTION** Since  $125 = 5^3$ , we have  $5^{2x-1} = 5^3$ . The base **5** is positive and is not **1**, so the exponential function  $5^t$  is one-to-one and  $2x - 1 = 3$ . Hence  $x = 2$ .

### 4 CHANGE OF BASE

Use the change-of-base formula to calculate  $\log_4 32$ . Show the formula you use and give the exact answer.

**SOLUTION**

$$\log_4 32 = \frac{\ln 32}{\ln 4} = \frac{\ln(2^5)}{\ln(2^2)} = \frac{5 \ln 2}{2 \ln 2} = \frac{5}{2}.$$

The formula is valid because the input **32** is positive and the bases **4** and **e** are positive and not equal to **1**.

## Part B · Reasoning and modelling

### 5 LOGARITHMIC EQUATION

Solve over the real numbers:

$$\log_2(x - 1) + \log_2(x + 1) = 3.$$

State the domain before combining the logarithms. Check every possible answer in the original equation.

**SOLUTION** Both logarithms require positive inputs:  $x - 1 > 0$  and  $x + 1 > 0$ . Therefore  $x > 1$ . On this domain, the product law gives

$$\log_2((x - 1)(x + 1)) = 3 \iff x^2 - 1 = 8 \iff x^2 = 9.$$

The algebra gives  $x = 3$  or  $x = -3$ , but the domain keeps only  $x = 3$ . Check:  $\log_2(2) + \log_2(4) = 1 + 2 = 3$ .

## 6 GROWTH COMPARISON

Put the following functions in order from slowest growth to fastest growth as the positive integer  $n$  becomes very large:

$$\log_2 n, \quad n, \quad n \log_2 n, \quad n^2, \quad 2^n.$$

In two or three sentences, explain why checking one value such as  $n = 3$  does not prove the full ranking.

**SOLUTION** The eventual order is

$$\log_2 n < n < n \log_2 n < n^2 < 2^n.$$

This is an eventual statement: it becomes true once  $n$  is large enough, not necessarily at every positive integer. For example, at  $n = 3$ ,  $2^n = 8 < 9 = n^2$ , but for sufficiently large  $n$  the exponential  $2^n$  is larger than  $n^2$ .

## 7 PERIODIC MODEL

Let  $t$  be the number of hours after midnight. A simple model for temperature is

$$T(t) = 3 \sin\left(\frac{\pi}{6}(t - 1)\right) + 20 \quad (0 \leq t \leq 12).$$

- State the amplitude, midline, period, and range.
- At what time does the model first reach a maximum? At what time does it reach a minimum?
- Explain why  $(t - 1)$  moves the graph right by 1 hour, rather than left by 1 hour.

**SOLUTION** The amplitude is 3, the midline is  $y = 20$ , the period is  $\frac{2\pi}{\pi/6} = 12$  hours, and the range is  $[17, 23]$ . A maximum occurs when the sine input is  $\pi/2$ , so  $\frac{\pi}{6}(t - 1) = \frac{\pi}{2}$  gives  $t = 4$ . A minimum occurs when the input is  $3\pi/2$ , giving  $t = 10$ . In  $f(t - 1)$ , the new input must be  $t = 1$  to give the old value  $f(0)$ , so the graph moves right by 1.

## Part C · Communicate precisely

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### 8 FIND THE ERROR

A student writes: “ $\log_2(u + v) = \log_2 u + \log_2 v$  because logarithms turn an operation into addition.”

- Identify the false statement.
- Give a counterexample with positive  $u$  and  $v$ .
- Write the correct logarithm law for a product, including its condition.

**SOLUTION** The false statement is  $\log_2(u + v) = \log_2 u + \log_2 v$ . Take  $u = v = 1$ :  $\log_2(1 + 1) = \log_2 2 = 1$ , while  $\log_2 1 + \log_2 1 = 0 + 0 = 0$ . The correct product law is  $\log_2(uv) = \log_2 u + \log_2 v$  for  $u > 0$  and  $v > 0$ .

### 9 MAKE A CLAIM PRECISE

The statement “An exponential function is always bigger than a polynomial” is false or unclear as written. Rewrite it as a true statement about  $c^n$  and  $n^k$ , including all important conditions. Then give one numerical example showing why the words “for sufficiently large  $n$ ” are necessary.

**SOLUTION** One correct version is: “For every fixed  $c > 1$  and  $k > 0$ , there is a positive integer  $N$  such that  $c^n > n^k$  for every integer  $n \geq N$ .” The word “fixed” matters: the bases and powers are not changing with  $n$ . For  $c = 2$ ,  $k = 2$ , and  $n = 3$ , we have  $2^3 = 8 < 9 = 3^2$ , so the inequality is not true for every positive  $n$ .